

Victor Diab

Portfolio: victordiab.com | LinkedIn: linkedin.com/in/victordiab
Phone: +1 (802) 330-1610 | Email: victorpdia@gmail.com

Skills

Software

- Unity 6
- Unreal Engine 5 (UE5)
- RenderDoc | Nvidia Nsight
- Jira | Azure

Programming Methodologies

- C++ | C# | C | Python
- GLSL | HLSL
- Vulkan | OpenGL
- Git | SVN

Mathematical Methodologies

- Linear Algebra
- Vector Calculus
- Numerical Methods
- LaTeX | MatLab

Education

Bachelor of Science in Game Programming & Bachelor of Science in Applied Mathematics

Champlain College in Burlington, VT | Completion Date: May 2026

GPA 3.49 | Dean's List (3.5+ GPA) Spring 2024, Fall 2024, and Spring 2025

Work Experience

Game Programming & Math Tutor | *Champlain College* | Jan 2025 - Jan 2026

- Provided support and detailed explanations to students for programming and mathematical topics such as graphics programming, game architecture, linear algebra, and vector calculus

Summer Programming Intern | *Maysalward* | Jun 2025 - Aug 2025

- Concepted, Designed, and implemented a mobile game prototype with professional feedback
- Researched and utilized design patterns to match features with known implementation strategies allowing for more scalable code
- Utilized technologies including mobile development, XR development, and online networking

Summer Programming Intern | *Mental* | Jul 2024 - Aug 2024

- Implemented UI and audio passes for learning modules using the existing infrastructure in an educational game
- Fixed bugs found in testing as prioritized in sprint planning
- Created documentation for proprietary systems to allow for easier reference

Project Experience

Programmer | Raytraced Software Renderer (C++) | Feb 2026

- Created a raytraced software renderer with different scenes and texture mapping
- Rendered different geometry (Plane, Sphere, Cylinder, Capsule)
- Created different tools to improve the workflow and debugging (multithreading, normal texture, depth texture)

Programmer | Vulkan Renderer (C) | Jan 2026

- Implemented architecture for a vulkan renderer in the C language
- Observed the differences between modern graphics APIs and legacy APIs

Programmer | GPU Flocking (Unity, HLSL, C#) | Apr 2025

- Implemented the flocking algorithm such that it calculates boids on the GPU using compute shaders to save performance
- Optimized performance to simulate the movement of thousands of boids simultaneously while maintaining a stable 60 FPS
- Implemented instanced rendering of boids so they work with vertex shader animations created in shader graph

Lead Programmer & Graphics Programmer | *Cthulhu and Me* (Unity, HLSL, ShaderGraph, C#) | Jan 2025 - Apr 2025

- Worked with a team of 12 developers as the lead programmer ensuring quality control of any code in the game
- Conducted performance tests to ensure the game ran smoothly on target hardware
- Created multiple graphics effects utilizing shaders

Other Achievements

Scrum Master Certification | Mar 2025

- Practiced methods of team self organisation in a course on Agile workflow and Scrum in Game Development
- Demonstrated knowledge gained in certification exam

Hudson River Undergraduate Math Conference | Apr 2025

- An Exploration of Spline Applications

Hudson River Undergraduate Math Conference | Apr 2024

- Multi-attribute decision making for location optimization

Eagle Scout | May 2019

References are available upon request